

CAPSTONE PROJECT

AI TRAVEL PLANNER FOR STUDENTS

Presented By:

1. Student Name- TRIJIT ROY
2. College / Institution Name – JIS COLLEGE OF ENGINEERING
3. Department- Computer Application
4. AICTE ID- STU6a041b40a32621778654016

OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

Planning a trip is time-consuming and expensive for students on tight budgets, and today's travel apps don't solve the parts of the problem that matter most to them:

- Hard budget caps are ignored: apps display prices but never enforce a spending ceiling, so users overspend without realizing it.
- Group costs aren't itemized or split, forcing manual reconciliation after every trip.
- Student-specific savings, such as ID discounts, hostel rates, and off-peak fares, are invisible in mainstream planning tools.
- Manual planning takes hours across many tabs comparing prices, routes, and reviews, with no single source of truth.
- Net effect: students either overspend, under-plan, or skip trips altogether because the planning overhead outweighs the reward.

PROPOSED SOLUTION

- The proposed system turns a student's budget cap and interests into a cost-itemized, day-by-day itinerary that never exceeds that budget. The solution consists of the following components:
- Data Collection:
 - Gather points of interest, transit fares, and accommodation prices from OpenStreetMap, the Overpass API, and city transit data.
 - Pull a curated student-discount corpus (ISIC-accepted venues, hostel rates, museum and rail concessions).
- AI Orchestration:
 - Use an LLM to group candidate points of interest into themed daily plans based on the traveler's interests.
 - Retrieve relevant student discounts via a RAG lookup over the discount corpus.
- Budget Optimization:
 - Apply a 0/1 knapsack solver to pick the subset of each day's candidates that maximizes value without exceeding the day's budget slice.
 - Order the chosen stops with a nearest-neighbor TSP approximation to minimize travel time.
- Deployment:
 - Ship a Streamlit prototype (Next.js/React for production) with budget sliders, a map view, and a day-by-day itinerary timeline.
 - Deploy on Streamlit Community Cloud / Hugging Face Spaces for the hackathon demo.
- Evaluation:
 - Validate that 100% of generated itineraries respect the budget cap using the automated cost validator.
 - Measure average time-to-plan and cost savings versus manual planning.

SYSTEM APPROACH

The system approach outlines the strategy for building a budget-first, AI-orchestrated travel planner for students.

Key components:

- System requirements: FastAPI backend, Streamlit/React frontend, vector store for the discount RAG corpus, and OSM/Overpass API access for POI data
- Core libraries: LangChain + Anthropic SDK (AI orchestration), Pydantic (schema validation), Folium/Geopy (mapping & geocoding)
- Optimization libraries: pure Python for the knapsack solver and nearest-neighbor TSP approximation (no heavy solver dependency)

ALGORITHM & DEPLOYMENT

- The budget guarantee is enforced by classical algorithms, not the LLM. Here's the structure:
- **Algorithm Selection:**
 - A 0/1 knapsack solver picks the subset of candidate stops that maximizes total interest-match value without exceeding the budget — chosen because it gives a hard, provable spending guarantee, unlike an LLM's free-text estimate.
- **Data Input:**
 - Candidate points of interest with cost and a 0–1 value score (interest match × popularity), plus pairwise distances for route ordering.
- **Solving Process:**
 - Dynamic programming knapsack (discretized to ₹5 cost units) maximizes value per day; falls back to a greedy value/cost ratio for very large candidate sets to stay fast.
 - A nearest-neighbor TSP approximation then orders the chosen stops to minimize same-day backtracking.
- **Deployment:**
 - The optimizer and cost validator run server-side (FastAPI); the LLM call is stateless and swappable, with a deterministic offline fallback so the app works even without an API key.
 - The Streamlit prototype is deployable to Streamlit Community Cloud or Hugging Face Spaces in minutes.

RESULT

 **Welcome TRIJIT ROY!** VERIFIED

EduTravel Global Academic Travel Optimization Engine • JISCE

Log Out

1. Travel Scale

Travel Category

☐ Domestic (India) ☒ International (Global)

Global Destination

Tokyo (Japan) 

2. Trip Parameters

Semester Break Duration (Days)

3

Study Group Size

2  

3. Student Financials (INR Eqv.)

Max Budget Per Student (₹)

8000  

Vibe & Interests

Historical Study 

Nature & Relaxat... 

 

Generate Optimized Route for Tokyo (Japan)

VERIFIED GROUP EXPENDITURE

₹2,300

 Within strict JISCE student limits

ISIC / COLLEGE ID SAVINGS

₹300

 Saved on international entry fees

OPTIMIZED STUDY BREAK WAYPOINTS

4

 TSP transit routing active

Generated Itinerary — Tokyo (Japan)

 Day 1 Output — Cost: ₹0

Vibe: Post-exam relaxation and scenic outdoor retreats.

Global Geospatial Routing

 



edunet
foundation

CONCLUSION

- The AI Travel Planner shows that combining an LLM's taste for interest-matching with classical optimization (knapsack + nearest-neighbor TSP) gives students a planner that is both personalized and provably budget-safe. The main implementation challenge was balancing candidate diversity against solve speed; the fallback greedy selector keeps the app fast even for large POI sets. Since the budget guarantee is the core promise to students, accurate cost data and a strict validator matter more than raw model creativity.

FUTURE SCOPE

- Now: core planner shipping — budget optimizer, map UI, and student-discount RAG. Next: real-time disruption re-routing for delays, closures, and weather. Later: social group sync with collaborative editing and live group voting on itinerary changes. Future: multi-city rail-pass optimization and carbon-conscious routing across an entire semester's travel.

REFERENCES

- Sources and tools instrumental in developing this solution, including the mapping data provider, the AI orchestration framework, and the LLM API documentation:
- [LangChain — AI orchestration & prompt-chaining framework \(python.langchain.com\)](https://python.langchain.com)
- [Overpass API — OpenStreetMap query language for budget POI data](#)
- [Anthropic Claude — API & Messages documentation](#)

DEPLOYMENT LINK / GITHUB LINK

- Github Link: <https://github.com/trijitroy2006/EduTravel-AI-Planner>



THANK YOU